Project Title:

Auto MPG Performance Dashboard using Python Dash

Tools / Libraries Used:

- Python
- Dash
- Plotly Express
- Pandas
- HTML Components (Dash)

Project Description:

This project presents a **static data visualization dashboard for analyzing automobile fuel efficiency** using the **Auto MPG dataset**. The dashboard is developed with **Python Dash and Plotly** to display key performance indicators such as **average MPG, year-over-year fuel efficiency growth, average horsepower, and total number of vehicles**. Multiple visualizations including **MPG trends over years, fuel efficiency by vehicle origin, relationships between MPG and vehicle weight, MPG vs horsepower, and MPG distribution** help users understand performance patterns and factors affecting fuel efficiency. The dashboard demonstrates **data preprocessing, KPI calculation, and business-style data visualization using Python-based analytics tools**.

Code Implementation:

```
import dash

from dash import dcc, html

import plotly.express as px

import pandas as pd


# Load Dataset

columns = ["mpg","cylinders","displacement","horsepower",

        "weight","acceleration","model_year","origin","car_name"]


df = pd.read_csv("auto-mpg.data",

            names=columns,
```

```
              sep=r"\s+",
              na_values="?")

df.dropna(inplace=True)

origin_map = {1: "USA", 2: "Europe", 3: "Japan"}
df["origin"] = df["origin"].map(origin_map)

# KPI Calculations
avg_mpg = round(df["mpg"].mean(), 2)
avg_hp = round(df["horsepower"].mean(), 2)
total_vehicles = df.shape[0]

mpg_year = df.groupby("model_year")["mpg"].mean().reset_index()
yoy_growth = round(((mpg_year["mpg"].iloc[-1] - mpg_year["mpg"].iloc[0])
                / mpg_year["mpg"].iloc[0]) * 100, 2)

# Create Figures
fig_trend = px.line(mpg_year,
                x="model_year",
                y="mpg",
                title="MPG Trend Over Years")

fig_origin = px.bar(df,
                x="origin",
                y="mpg",
                title="Average MPG by Origin",
                barmode="group")

fig_weight = px.scatter(df,
                  x="weight",
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```
                    y="mpg",
                    title="MPG vs Weight")

fig_hp = px.scatter(df,
                  x="horsepower",
                  y="mpg",
                  title="MPG vs Horsepower")

fig_hist = px.histogram(df,
                    x="mpg",
                    nbins=15,
                    title="MPG Distribution")

# Initialize App
app = dash.Dash(__name__)
app.title = "Auto MPG Dashboard"

# Layout
app.layout = html.Div([

    html.H1("Auto MPG Performance Dashboard",
            style={"textAlign": "center"}),



    # KPI SECTION
    html.Div([

        html.Div([
            html.H3("Avg MPG"),
            html.H2(avg_mpg)
        ], className="card"),
```

```
    html.Div([
        html.H3("YoY Growth"),
        html.H2(f"{yoy_growth}%")
    ], className="card"),

    html.Div([
        html.H3("Avg Horsepower"),
        html.H2(avg_hp)
    ], className="card"),

    html.Div([
        html.H3("Total Vehicles"),
        html.H2(total_vehicles)
    ], className="card"),

], style={"display": "flex",
          "justifyContent": "space-around",
          "marginBottom": "30px"}),


# MIDDLE SECTION
html.Div([
    dcc.Graph(figure=fig_trend, style={"width": "65%"}),
    dcc.Graph(figure=fig_origin, style={"width": "35%"})
], style={"display": "flex"}),


# BOTTOM SECTION
html.Div([
    dcc.Graph(figure=fig_weight, style={"width": "33%"}),
```

```
        dcc.Graph(figure=fig_hp, style={"width": "33%"}),
        dcc.Graph(figure=fig_hist, style={"width": "33%"})
    ], style={"display": "flex"})

])
# Run Server
if __name__ == "__main__":
    app.run(debug=True)
```

Output: